🏦

PRD – Proactive Financial Stress Detection | Banking App

Status	In progress
Type	PRD

PRD – Proactive Financial Stress Detection

Banking App (Neobank: Jupiter / Revolut-style)

> **Document Status:** Draft

> **Author:** Monika M

> **Date:** July 2026

> **Version:** v1.0

---

1. 📌 Problem Statement

A meaningful segment of neobank users experience **sudden financial stress** — irregular income gaps, overdrafts, and missed bill payments — but only reach out for help *after* the situation has escalated into late fees or credit damage.

The app currently **treats every user the same** regardless of their financial health signals, missing the opportunity to intervene early in a helpful, non-intrusive way.

> **How might we** proactively identify users showing early signs of financial stress and offer timely, relevant support — such as flexible payment options or budgeting nudges — while maintaining user trust and avoiding the perception of surveillance or judgment?

---

2. 🎯 Goals & Non-Goals

Goals

- Proactively detect early financial stress signals using in-app behavioral data
- Surface timely, empathetic nudges and flexible support options *before* a crisis
- Build user trust through transparency, consent, and tone
- Improve financial outcomes for at-risk users

Non-Goals

- Replacing human financial advisors or customer support
- Accessing third-party financial data without explicit user consent
- Offering full-scale credit or lending products (out of scope for v1)
- Affecting the core banking/payments UX for non-stressed users

---

3. 👥 User Personas

Persona 1 – Rahul, 24 | Gig Worker

- Freelance designer with irregular monthly income
- Misses bill payments in low-income months
- Feels shame asking for help; avoids banking apps when stressed
- **Need:** Non-judgmental early warning + short-term liquidity support

Persona 2 – Priya, 27 | First-Time Earner

- Just started her first job; no prior financial planning habits
- Overspends early in the month and runs out before salary
- **Need:** Budgeting nudges and spending insights before it's too late

Persona 3 – Arun, 35 | Mid-Career Professional

- Stable income but hit a sudden expense (medical/car repair)
- Usually reliable but temporarily cash-strapped
- **Need:** Flexible bill payment options and a short grace period

---

4. 🔍 User Research Insights

- **68%** of neobank users who missed a bill payment showed detectable stress signals 5–7 days prior (low balance + delayed transactions)
- **54%** of users said they'd appreciate proactive help if it felt like "a trusted friend, not a bank"
- **Top fear:** Being judged or having their credit score flagged for "being in trouble"
- **Preferred tone:** Friendly, factual, and optional — not alarmist or pushy

---

5. 📐 User Stories

Must Have

- As a user with a low balance approaching a bill due date, I want to receive a gentle nudge so I can take action before missing the payment.
- As a gig worker, I want to access a salary advance or short payment deferral so I can cover urgent bills without overdrafting.
- As a user, I want to opt in or out of financial health monitoring so I feel in control of my data.

Should Have

- As a user, I want to see a simple weekly spending summary so I can course-correct mid-month.
- As a user, I want to set a personal "safety net" balance so the app alerts me before I go below it.

Could Have

- As a user, I want to chat with a financial wellness bot for quick budgeting advice.
- As a user, I want to see my financial health score trend over 3 months.

Won't Have (v1)

- Full credit scoring integration
- Third-party account aggregation (Open Banking)
- Investment or savings product recommendations

---

6. 🚦 Stress Signal Detection Logic

⚠️ **Privacy-first principle:** All signal detection uses only in-app behavioral data. No third-party data. Users are informed and opt-in to Financial Health features.

Tier 1 – Mild Stress Signals

- Balance drops below 20% of average monthly credit within first 15 days
- Bill payment delayed by 3–5 days vs. user's typical pattern
- Multiple small peer-to-peer transfers in a short window (borrowing pattern)

Tier 2 – Moderate Stress Signals

- Balance drops below ₹500 with a bill due in < 5 days
- Two or more consecutive months with late bill payments
- Repeated failed auto-debit attempts

Tier 3 – High Stress Signals

- Overdraft triggered
- Bill payment missed entirely
- No transactions for 7+ days (disengagement signal)

---

7. 💡 Feature Set & Prioritization (RICE)

Feature	Reach	Impact	Confidence	Effort	RICE Score	Priority
Smart Balance Nudge	10,000	2	80%	2	8,000	🟢 Must Have
Bill Payment Flexibility (3-	8,000	2	75%	3	4,000	🟢 Must Have

Feature	Reach	Impact	Confidence	Effort	RICE Score	Priority
day grace)						
Salary Advance (Micro-credit)	6,000	3	70%	4	3,150	🟡 Should Have
Weekly Spending Summary	12,000	1	85%	2	5,100	🟡 Should Have
Safety Net Balance Alert	9,000	1	80%	1	7,200	🟡 Should Have
Financial Wellness Chatbot	5,000	1	60%	5	600	🔵 Could Have

8. 🗺️ User Flow – Smart Balance Nudge (MVP)

```
User opens app OR background trigger fires
        ↓
Stress signal detected (Tier 1 or 2)
        ↓
System checks: Has user opted into Financial Health?
   ├── NO → No action (respect preference)
   └── YES ↓
Nudge generated (friendly, contextual)
        ↓
Push notification OR in-app banner shown
        ↓
User taps nudge → lands on Financial Health card
        ↓
   Options shown:
   ├── View spending breakdown
   ├── Request bill grace period
   ├── Explore salary advance
   └── Dismiss (no judgment)
```

9. 🎨 UX & Design Principles

Tone of Voice

- ✅ **Do:** "Hey Monika, your rent is due in 3 days and your balance looks a little tight. Want to explore options?"
- ❌ **Don't:** "Warning: You may be financially stressed. Take action now."

Design Guardrails

- **Opt-in only** — Financial Health monitoring is off by default
- **Non-intrusive** — Max 1 nudge per stress episode, not daily alerts

- **Easy exit** — One tap to dismiss, no guilt messaging
- **Transparent** — Always show *why* the nudge appeared ("Based on your balance and upcoming bills")
- **No shame language** — Never use words like "struggling," "overdue," or "failed"

---

10. 🔒 Privacy & Trust Framework

Concern	Our Approach
Surveillance perception	Only use in-app data; clearly disclosed in onboarding
Data misuse	Stress signals never shared with third parties or credit bureaus
User control	Full opt-out available at any time in Settings
Transparency	"Why am I seeing this?" link on every nudge
Tone	Empathetic, optional, never alarmist

---

11. 📊 Success Metrics

North Star Metric

> % of Tier 2 stress users who take a proactive action **before** a missed payment

Primary Metrics

- **Stress-to-action rate:** % of nudged users who engage with a support feature
- **Late payment reduction:** % decrease in missed bill payments among opted-in users
- **Salary advance adoption:** # of users using micro-advance feature per month

Secondary Metrics

- **Nudge CTR:** Click-through rate on Financial Health nudges (target: >25%)
- **Feature engagement rate:** % of opted-in users who return to Financial Health tab weekly

Guardrail Metrics

- **Opt-out rate:** Must stay below 15% (signals the nudges feel intrusive)
- **Negative feedback rate:** App store reviews mentioning "surveillance" or "creepy" — target: 0 increase
- **Support tickets from nudged users:** Should not increase

---

12. 🗓️ Go-To-Market & Launch Plan

Phase 1 – Alpha (Weeks 1–4)

- Internal dogfooding with 100 employees
- Validate stress signal accuracy with data science team
- Refine nudge copy with UX writers

Phase 2 – Closed Beta (Weeks 5–10)

- Opt-in rollout to 5,000 users (high-stress cohort identified by data team)
- A/B test nudge formats: push notification vs. in-app banner
- Weekly feedback surveys

Phase 3 – Gradual Rollout (Weeks 11–16)

- Expand to 20% of user base with opt-in toggle in app
- Monitor guardrail metrics closely
- PR angle: "Your bank that looks out for you"

Phase 4 – Full Launch

- Feature flagged on for all users (opt-in remains)
- App store feature pitch to Apple/Google
- Partnership with financial wellness content creators

---

13. ⚠️ Risks & Mitigations

Risk	Likelihood	Impact	Mitigation
Users feel surveilled	Medium	High	Opt-in design + transparent "why" messaging
False positives (wrong users flagged)	Medium	Medium	Threshold tuning in beta; user feedback loop
Low opt-in rate	Medium	High	Compelling onboarding value prop for Financial Health
Regulatory concerns (RBI/DPDP Act)	Low	High	Legal review before launch; data minimization by design
Feature misuse by bad actors	Low	Medium	Rate limiting on advance requests; fraud detection layer

---

14. 🔗 Dependencies

- **Data Science team:** Stress signal model training and validation
- **Backend/Infra:** Real-time balance monitoring pipeline
- **Legal & Compliance:** DPDP Act compliance review, RBI guidelines for micro-credit
- **UX/Design:** Nudge design system, Financial Health tab UI
- **Content/Copy:** Empathetic tone guide for all financial stress messaging
- **Partnerships (v2):** NBFC partner for salary advance product

---

15. 📎 Open Questions

- [ ] What is the exact balance threshold for Tier 1 vs. Tier 2 detection? (Needs data science input)
- [ ] Should salary advance be built in-house or via an NBFC API partnership?
- [ ] How do we handle users who opt out and then hit Tier 3 stress? (Any safety net?)

- [ ] What's the maximum number of nudges per month to avoid fatigue?
- [ ] How do we measure long-term financial health improvement (6-month cohort study)?

---

16. ✅ Approval & Sign-Off

Role	Name	Status	Date
Product Manager	Monika M	✍️ Draft	July 2026
Engineering Lead	TBD	⏳ Pending	—
Design Lead	TBD	⏳ Pending	—
Legal & Compliance	TBD	⏳ Pending	—